

Assignment

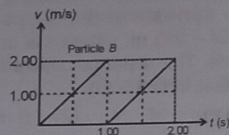
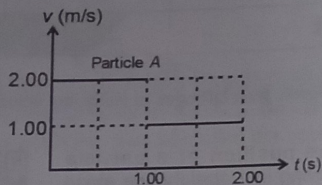
QR codes are provided for Video solutions of the questions. Please scan QR code to watch the solution.

SECTION - A

Objective Type Questions (One option is correct)

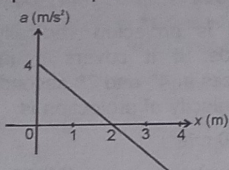
1. A particle moving in a straight line covers the first half of distance with speed of 12 m/s and the other half of the distance in two equal time intervals with speed of 18 m/s and 30 m/s respectively. The average speed of particle during motion is
 (1) 16 m/s (2) 24 m/s
 (3) 18 m/s (4) 12 m/s
2. The position x of a particle moving in a straight line along x -axis varies with time t as $x = (t - 3)^2$, where x is in metres and t is in seconds. The displacement of particle till it comes to rest is
 (1) -9 m (2) -5 m
 (3) 12 m (4) 0
3. A particle is at rest at $t = 0$. If acceleration of the particle is given as $a = \sin \pi t + \cos \pi t$, in SI units, then the maximum speed of particle is
 (1) $\frac{2}{\pi}$ m/s (2) $\frac{2\sqrt{2}}{\pi}$ m/s
 (3) $\frac{(\sqrt{2} + 1)}{\pi}$ m/s (4) $\frac{4\sqrt{2}}{\pi}$ m/s
4. A particle moves in a straight line such that its acceleration a varies with speed v as $a = -2\sqrt{v}$ m/s², where v is measured in m/s. If initial velocity is 4 m/s, then time taken by the particle to come to stop is
 (1) $\sqrt{2}$ s (2) 2 s
 (3) $2\sqrt{2}$ s (4) 4 s
5. The position of a particle moving in a straight line along x -axis varies with time t as $x = (12t - 3t^2)$ m, where t is in seconds. The distance covered by particle in first 5 s is
 (1) 33 m (2) 15 m
 (3) 45 m (4) 39 m
6. A ball is dropped from rest at a height h from the ground. If it travels a distance of $\frac{9h}{25}$ in its last second of motion, then value of h is ($g = 10$ m/s²)
 (1) 225 m (2) 125 m
 (3) 150 m (4) 80 m
7. A ball is projected vertically upwards. If it covers equal distances in 4th and 7th second, then velocity of projection is ($g = 10$ m/s²)
 (1) 20 m/s (2) 50 m/s
 (3) 60 m/s (4) 30 m/s
8. From a height, two balls are thrown with equal speed, one vertically upwards and other vertically downwards. They take time t_1 and t_2 respectively to reach the ground. If a third ball is dropped from same height, then time taken to reach the ground will be
 (1) $\frac{t_1 + t_2}{2}$ (2) $\sqrt{t_1 t_2}$
 (3) $\frac{2t_1 t_2}{t_1 + t_2}$ (4) $\frac{\sqrt{t_1^2 + t_2^2}}{2}$
9. Two particles A and B start from the same point and move in the positive x -direction. In a time interval of 2.00 s after they start, their velocities v vary with time t as shown in the following figures. What is the maximum separation between the particles during this interval?

76 Motion in a Straight Line



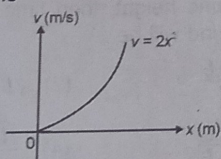
- (1) 1.00 m (2) 1.25 m
(3) 1.50 m (4) 2.00 m

10. The acceleration-position graph of a particle moving in a straight line along x-axis is shown in figure. The particle starts from rest at origin. The maximum speed of particle is



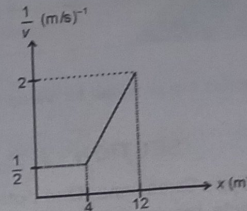
- (1) 2 m/s (2) $2\sqrt{2}$ m/s
(3) 4 m/s (4) $4\sqrt{2}$ m/s

11. The velocity-position graph of a particle moving in a straight line along x-axis is given below. Acceleration of particle at $x = 2$ m is



- (1) 8 m/s^2 (2) 16 m/s^2
(3) 64 m/s^2 (4) 32 m/s^2

12. Graph of $\left(\frac{1}{v}\right)$ versus x for a particle under motion is shown in figure, where v is velocity and x is position. Find the time taken by particle to move from $x = 4$ m to $x = 12$ m.



- (1) 8 s (2) 10 s
(3) 12 s (4) 6 s
13. Two cars are moving towards each other on a straight road with accelerations 5 m/s^2 and 15 m/s^2 . At a certain instant, their separation is 400 m and their velocities are 10 m/s and 20 m/s. How long will it take for the cars to collide?

- (1) 12 s (2) $(2 + \sqrt{3})$ s
(3) 5 s (4) $(1 + \sqrt{5})$ s

14. A boat goes from point A to B downstream of a river flowing with speed 6 km/h and then comes back to point A. If time taken in upstream journey is double of that in downstream journey then speed of boat in still water is

- (1) 6 km/h (2) 18 km/h
(3) 12 km/h (4) 24 km/h

15. A large wooden raft is flowing with a river stream. A man on the raft dives into river and swims down the stream for 15 minutes and then he turns about and comes back to the raft. The swimming speed of man is 5 km/h in still water and the river is flowing with speed 4 km/h. The distance moved by the raft till the man comes back to raft is

- (1) 1 km (2) 2 km
(3) 3 km (4) 4 km

SECTION - B

Motion in a Straight Line 77

Objective Type Questions

(One or more than one option(s) is/are correct)

1. Select the correct statements. For a particle moving on a straight line.

- (1) Average speed is always equal to magnitude of average velocity
- (2) Average speed may be greater than magnitude of average velocity
- (3) Average velocity = instantaneous velocity if velocity is constant
- (4) Moving with constant acceleration, average velocity for a given time interval is arithmetic mean of initial and final velocity

2. The displacement (x) of a particle moving along x -axis varies with time (t) as $x = (2t^2 - 3t^3)$ m, where t is in seconds, then

- (1) The particle returns to its initial position after $\frac{3}{2}$ s
- (2) The particle comes to rest at $t = \frac{4}{9}$ s
- (3) Initial velocity of particle is zero
- (4) Maximum velocity in positive x -direction is $\frac{4}{9}$ m/s

3. The motion of a particle moving along x -axis is represented by the equation $\frac{dv}{dt} = 6 - 3v$, where v is in m/s and t is in second. If the particle is at rest at $t = 0$, then

- (1) The speed of the particle is 2 m/s when the acceleration of particle is zero
- (2) After a long time the particle moves with a constant velocity of 2 m/s
- (3) The speed is 0.5 m/s, when the acceleration is half of its initial value
- (4) The magnitude of final acceleration is 6 m/s^2

4. A particle starts from rest and moves with acceleration $a = 4 \sin\left(\frac{\pi t}{2}\right) \text{ m/s}^2$ where t is in seconds. Initially the particle is at origin and moves along x -axis, then

- (1) Maximum speed of the particle is $\frac{16}{\pi} \text{ m/s}$
- (2) Direction of acceleration changes alternatively
- (3) Direction of velocity changes alternatively
- (4) Speed of the particle is $\frac{8}{\pi} \text{ m/s}$ at $t = 1 \text{ s}$

5. A particle moves in a straight line with initial velocity u_0 . Retardation acting on the particle is kv , where k is a positive constant and v is instantaneous velocity. Then

- (1) $v(t) = u_0 e^{-kt}$
- (2) $v(t) = u_0 (1 - e^{-kt})$
- (3) Displacement of particle at time t is $\frac{u_0}{k} e^{-kt}$
- (4) Displacement of particle when it comes to rest is $\frac{u_0}{k}$

6. The acceleration (a) of a particle moving in a straight line varies with velocity (v) as $a = (1 - v) \text{ m/s}^2$, where v is in m/s. If initially the particle was at rest, then

- (1) Maximum velocity of the particle will be 1 m/s
- (2) Maximum acceleration is 1 m/s^2
- (3) The particle will finally come to rest
- (4) Velocity of particle at $t = \ln 2 \text{ s}$ is $\frac{1}{2} \text{ m/s}$

7. Select the correct formulae for uniformly accelerated motion along a straight line.

- (1) $v = u + at$
- (2) $s = \left(\frac{u+v}{2}\right)t$
- (3) $s = vt - \frac{1}{2}at^2$
- (4) $s = \frac{v^2 - u^2}{2a}$

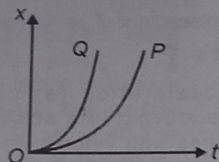
8. A ball is projected vertically up with speed 20 m/s. Take $g = 10 \text{ m/s}^2$

- (1) It covers a distance of 5 m in 2nd second of its motion
- (2) The displacement in 2nd and 3rd second are equal
- (3) The distance covered in 2nd and 3rd second are equal
- (4) The average speed for first 4 seconds is zero

9. A ball is projected vertically upwards from ground with velocity $v = 24 \text{ m/s}$. Assume the air resistance produces a constant retardation of 2 m/s^2 . Then ($g = 10 \text{ m/s}^2$)

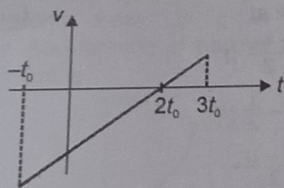
- (1) The maximum height attained is 28.8 m
- (2) Time of ascent is more than time of descent
- (3) Time of descent is more than time of ascent
- (4) Velocity of particle just before it hits the ground is $8\sqrt{6} \text{ m/s}$

10. The position-time graph for two particles P and Q moving on x-axis is shown in the figure. Then



- (1) The separation between P and Q remains constant with time
- (2) The acceleration of Q is greater than that of P
- (3) The relative velocity of Q w.r.t. P is increasing with time
- (4) The relative velocity of Q w.r.t. P is decreasing with time

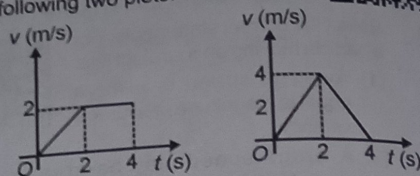
11. Figure shows the velocity of a particle moving on x-axis.



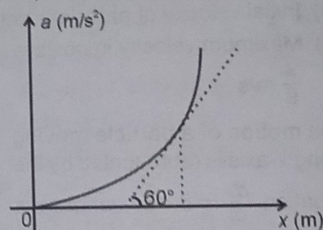
Which of the following statements is correct for the interval shown in graph?

- (1) Speed of particle is increasing continuously
- (2) The particle changes its direction of motion at some point of time
- (3) The particle does not return to its starting point at the end of the interval
- (4) The acceleration of particle changes from -ve to +ve

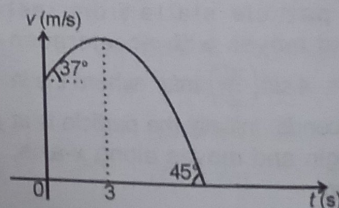
12. The velocity versus time graph of two particles moving along x-axis varies as shown in the following two plots. Then



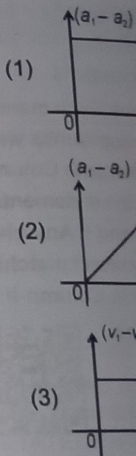
- (1) Maximum separation between two particle is 2 m
 - (2) Maximum separation between the two particles is 3 m
 - (3) Maximum separation occurs at $t = 2 \text{ s}$
 - (4) Maximum separation occurs at $t = 3 \text{ s}$
13. A particle starts moving along x-axis from origin with initial velocity of 3 m/s . The acceleration a of the particle varies with position x as $a = Kx^2$, where K is a constant. At $x = \sqrt{3} \text{ m}$, the tangent to the graph makes an angle of 60° with x-axis. Then at $x = \sqrt{3} \text{ m}$ (v represents velocity of particle at position x)



- (1) $v = 3\sqrt{2} \text{ m/s}$
 - (2) $v = \sqrt{(9 + \sqrt{3})} \text{ m/s}$
 - (3) $a = 3 \text{ m/s}^2$
 - (4) $a = 1.5 \text{ m/s}^2$
14. A particle moves in a straight line under an acceleration varying linearly with time. Its velocity-time graph is shown in figure. Then

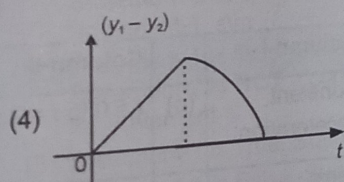
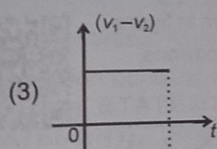
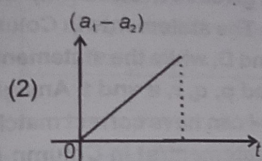
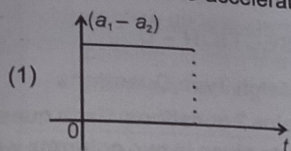


- (1) The particle comes to rest
 - (2) The initial velocity is 30 m/s
 - (3) Maximum velocity is 30 m/s
 - (4) The acceleration at $t = 3 \text{ s}$ is 30 m/s^2
15. Two stones are thrown upwards simultaneously from the same point with velocities u_1 and $u_2 = 30 \text{ m/s}$. The following curves show the relative velocity of the stones with time t . (A) $(u_1 - u_2) = \text{relative velocity}$ (B) $(u_1 + u_2) = \text{relative velocity}$ (C) $(u_1 - u_2) = \text{relative velocity}$ (D) $(u_1 + u_2) = \text{relative velocity}$



This section contains paragraphs, statements, and questions. Some of the questions are multiple choice, and some are numerical. The questions are linked to the paragraphs.

- (1) The particle comes to rest at $t = 7$ s
 (2) The initial velocity of particle is 1 m/s
 (3) Maximum velocity of particle is 2 m/s
 (4) The acceleration of particle is zero at $t = 3$ s
15. Two stones are thrown vertically upwards simultaneously from the same point on the ground with velocities $u_1 = 40$ m/s and $u_2 = 30$ m/s. Which of the following curve(s) represent(s) correct variation with time t ? (Assume that stone does not bounce back after collision with ground)
- ($y_1 - y_2$) = relative separation between stones
 ($v_1 - v_2$) = relative velocity of two stones
 ($a_1 - a_2$) = relative acceleration of two stones



SECTION - C

Linked Comprehension Type Questions

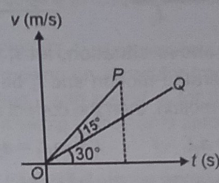
This section contains 2 paragraphs. Based upon each paragraph, some multiple choice questions have to be answered. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Comprehension-I

Graphs are very useful to represent a physical situation. Various quantities can be easily represented on graphs and other quantities can be determined from the graph. For example the slope of velocity-time graph represents instantaneous acceleration. For a motion with constant acceleration slope of velocity-time graph is constant. If acceleration is changing with time, slope with change and thus velocity-time graph will be a non-linear curve. Further the area of velocity-time graph gives displacement.

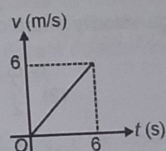
Choose the correct answer :

1. The velocity-time graph for two objects P and Q are shown. The ratio of their respective acceleration is



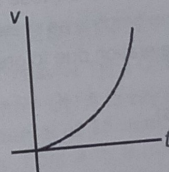
- (1) 1 (2) $\sqrt{3}$
 (3) $\frac{1}{\sqrt{3}}$ (4) 3

2. For the given velocity-time graph, the displacement of the object between $t = 2$ s to $t = 4$ s is



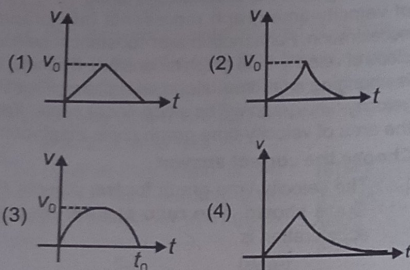
- (1) 6 m (2) 10 m
 (3) 18 m (4) 2 m

3. For the given velocity-time graph the correct acceleration-time relationship is



- (1) Acceleration is constant
 (2) Acceleration is increasing with time
 (3) Acceleration is decreasing with time
 (4) None of these

4. A car starts from rest with a constant acceleration a_1 for some time and attains a velocity v_0 . Then it immediately retards at a constant rate a_2 to rest. The total time of journey is t_0 . Identify the correct velocity-time graph.



5. For the above situation, let t_1 be the time of accelerated motion and t_2 be the time of retarded motion, then the correct relation is

(1) $a_1 t_1 = a_2 t_2$ (2) $a_1 t_1 = a_2 t_2^2$
 (3) $a_1 t_1^2 = a_2 t_2^2$ (4) $a_1 t_1^2 = a_2 t_2$

6. The correct relation between v_0 , a_1 , a_2 and t_0 is

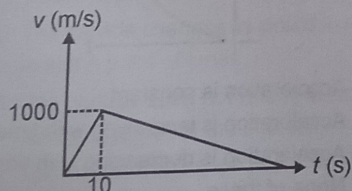
(1) $v_0 = \left(\frac{a_1 a_2}{a_1 - a_2} \right) t_0$ (2) $v_0 = \left(\frac{a_1 a_2}{a_1 + a_2} \right) t_0$
 (3) $v_0 = \left(\frac{a_1 + a_2}{2} \right) t_0$ (4) $v_0 = \sqrt{a_1 a_2} t_0$

7. The average velocity of the car for the entire trip is

(1) v_0 (2) $\frac{v_0}{2}$
 (3) $\frac{v_0}{3}$ (4) $\frac{v_0}{4}$

Comprehension-II

A rocket is fired upwards such that its engine takes 10 s to explode fully. Its velocity-time graph is as shown in the figure. Acceleration due to gravity is 10 m/s^2 .



Choose the correct answer :

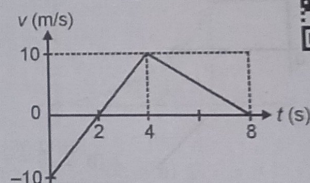
1. The total time for which the rocket moves upward is
 (1) 10 s (2) 100 s
 (3) 110 s (4) 20 s
2. The height reached by the rocket till its engine explodes is
 (1) 1000 m (2) 5000 m
 (3) 10,000 m (4) 20,000 m
3. The maximum height attained by the rocket is
 (1) 5000 m (2) 50,000 m
 (3) 55,000 m (4) 100,000 m

SECTION - D

Matrix-Match Type Questions

This section contains 2 questions. Each question contains statements given in two columns which have to be matched. The statements in Column-I are labelled A, B, C and D, while the statements in Column-II are labelled p, q, r, s and t. Any given statement in Column-I can have correct matching with One OR More statement(s) in Column-II.

1. For the v-t graph shown in figure match the following columns.



	Column-I	Column-II
(A)	Constant acceleration	(p) $t = 0$ to $t = 2$ s
(B)	Zero displacement	(q) $t = 2$ s to $t = 4$ s
(C)	Increasing speed	(r) $t = 0$ to $t = 4$ s
(D)	Negative displacement	(s) $t = 4$ s to $t = 8$ s

2. Two trains each 100 m are moving 54 km/h (for Train-1) and 36 km/h (for Train-2) on parallel tracks. Match the statements in column-I with the statements in column-II.

	Column-I
(A)	Time taken for the trains to pass each other (in s)
(B)	Distance travelled by 1 st train relative to 2 nd train (in m)
(C)	Distance travelled by 2 nd train relative to 1 st train (in m)
(D)	Displacement of the 1 st train w.r.t. the 2 nd train (in m)

Assertion-I

This section contains STATEMENT-1 and STATEMENT-2. Match the statements in column-I with the statements in column-II. Any given statement in column-I can have correct matching with One OR More statement(s) in column-II.

- (1) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation of Statement-1.
 (2) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation of Statement-1.
 (3) Statement-1 is true, Statement-2 is false.
 (4) Statement-1 is false, Statement-2 is true.

1. STATEMENT-1: The velocity of a particle moving in a circular path is constant. STATEMENT-2: The acceleration of a particle moving in a circular path is constant.

2. Two trains each of length 100 m are moving with speeds 54 km/h (for Train-1) and 90 km/h (for Train-2) on two parallel tracks. Match the entries given in column-I with the entries given in column-II.

Column-I	Column-II
(A) Time taken in passing each other (in s)	(p) 125
(B) Distance travelled by 1 st train during passing (in m)	(q) 500
(C) Distance travelled by 2 nd train during passing (in m) if both trains are moving in same direction	(r) 20
(D) Displacement of 1 st train w.r.t. 2 nd train in 2 s (in m)	(s) 300
	(t) 5

SECTION - E

Assertion-Reason Type Questions

This section contains 5 questions. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 - (2) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 - (3) Statement-1 is True, Statement-2 is False
 - (4) Statement-1 is False, Statement-2 is True
1. STATEMENT-1 : The instantaneous velocity is always tangential to the path.

and

STATEMENT-2 : The slope of position-time graph represents the velocity.

STATEMENT-1 : Zero velocity of a particle does not mean zero acceleration.

and

STATEMENT-2 : The moment at which velocity is zero, its rate of change may be non-zero.

3. STATEMENT-1 : Two bodies projected vertically upward with different initial speeds cover same distances in last second of their upward journey.

and

STATEMENT-2 : For uniformly accelerated motion, distance travelled in a time interval before coming to rest does not depend on initial speed.

4. STATEMENT-1 : Acceleration of a car moving on a circular track of radius r with speed v , for the person sitting in the car is zero.

and

STATEMENT-2 : In the frame of reference of a person sitting in a car, the car always remains at rest irrespective of its motion with respect to ground.

5. STATEMENT-1 : For an observer looking out through the window of a fast moving train, the nearby objects appear to move in the opposite direction to the train, while the distant objects appear to be stationary.

and

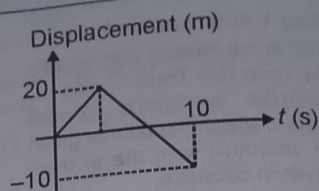
STATEMENT-2 : If the observer and the object are moving at velocities $\vec{V}_1$ and $\vec{V}_2$ respectively with reference to a laboratory frame, the velocity of the object with respect to the observer is $\vec{V}_2 - \vec{V}_1$.

SECTION - F

Integer Answer Type Questions

1. Buses ply from city A to city B at regular interval of 15 minutes. Every bus from A to B moves with constant speed 40 km/h. A car X moving from city B to city A meets buses at an interval of 5 minutes. The speed of the car X is $n \times 10$ km/h. Find the value of n .

2. The velocity of a particle moving on the x -axis is given by $v = x^2 + x$, where x is in m and v in m/s. What is its position (in m) when its acceleration is 30 m/s^2 ?
3. A parachutist falls freely from a helicopter for 10 s before the parachute open out. Then he descends with a net retardation of 2.5 ms^{-2} . If he bails out of the helicopter at a height of 2495 m and $g = 10 \text{ m/s}^2$, find his velocity on reaching the ground in m/s.
4. The diagram shows the displacement-time graph for the particle moving in a straight line. What is average speed (in m/s) for the interval 0-10 s?



5. A lift starts from rest at $t = 0$ and moves upwards with constant acceleration of 5 m/s^2 . At $t = 2 \text{ s}$, a bolt from the ceiling of lift detaches and falls freely. If height of lift is 7.5 m, then the bolt hits the floor of lift at $t = n \text{ s}$. Find the value of n . ($g = 10 \text{ m/s}^2$)

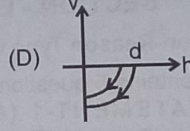
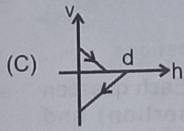
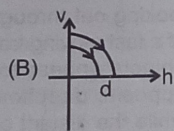
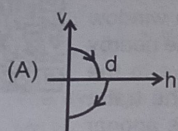


SECTION - G (PREVIOUS YEARS QUESTIONS)

Only One Option Correct Type Questions

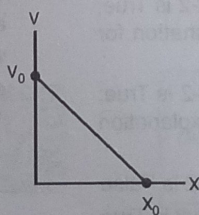
1. A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up vertically to a height $d/2$. Neglecting subsequent motion and air resistances, its velocity v varies with the height h above the ground as

(2000S)

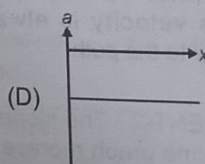
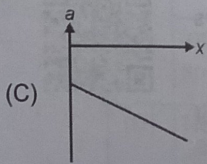
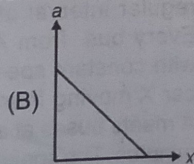
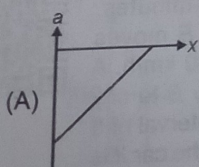


2. The velocity-displacement graph of a particle moving along a straight line is shown

(2005S)



The most suitable acceleration-displacement graph will be



One or More Option(s) Correct Type Questions

Motion in a Straight Line 83

3. A particle of mass m moves on the x -axis as follows: it starts from rest at $t = 0$ from the point $x = 0$, and comes to rest at $t = 1$ at the point $x = 1$. No other information is available about its motion at intermediate time ($0 < t < 1$). If α denotes the instantaneous acceleration of the particle, then

- (A) α cannot remain positive for all t in the interval $0 \leq t \leq 1$.
 (B) $|\alpha|$ cannot exceed 2 at any point in its path.
 (C) $|\alpha|$ must be ≥ 4 at some point or points in its path.
 (D) α must change sign during the motion, but no other assertion can be made with the information given.

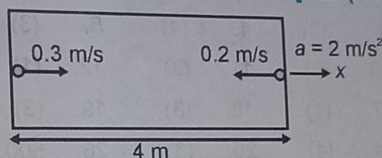
(1993)



Integer / Numerical Value Type Questions

4. A rocket is moving in a gravity free space with a constant acceleration of 2 m/s^2 along $+x$ direction (see figure). The length of a chamber inside the rocket is 4 m . A ball is thrown from the left end of the chamber in $+x$ direction with a speed of 0.3 m/s relative to the rocket. At the same time, another ball is thrown in $-x$ direction with a speed of 0.2 m/s from its right end relative to the rocket. The time in seconds when the two balls hit each other is

[JEE (Adv)-2014 (Paper-1)]



5. A solid horizontal surface is covered with a thin layer of oil. A rectangular block of mass $m = 0.4 \text{ kg}$ is at rest on this surface. An impulse of 1.0 Ns is applied to the block at time $t = 0$ so that it starts moving along the x -axis with a velocity $v(t) = v_0 e^{-t/\tau}$, where v_0 is a constant and $\tau = 4 \text{ s}$. The displacement of the block, in metres, at $t = \tau$ is _____. Take $e^{-1} = 0.37$.

[JEE (Adv)-2018 (Paper-2)]



